Ishan Deshpande

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EDUCATION

UT Austin

M.S. in Electrical and Computer Engineering (Spec. Comp. Arch. and Embedded Systems) **December 2026**
B.S. in Electrical and Computer Engineering **May 2025**
3.75 Cumulative GPA, 3.83 Graduate GPA
Relevant Coursework: Embedded Systems Lab, Digital Logic Design, ML/HW Codesign, Comp. Arch., Algorithms, Distributed Systems, Multithreaded Programming, RTOS, VLSI, Graphics, ML Systems, Compilers

EXPERIENCE

Applied Intuition **Feb 2027 - Present**
Core OS Engineer *Sunnyvale, CA*

Apple **June 2026 - Aug 2026**
Embedded Runtimes Intern *Cupertino, CA*

- Cut firmware memory footprint by shipping a lockless, interrupt-safe thread-pooling primitive for sharing OS thread stacks
- Quantified memory and performance tradeoffs across firmware subsystems, informing an adoption rollout plan

Meta Reality Labs **May 2025 - Aug 2025**
Codec Avatars Embedded Intern *Pittsburgh, PA*

- Accelerated sensor prototyping by building a custom assembler, emulator, and verifier for a programmable I/O state machine on FPGA, reducing development time and enabling flexible sensor integration.
- Improved model fidelity by integrating ultrasonic, radar, and IMU data to capture facial movements.

Tesla **May 2024 - Aug 2024**
Body Controls Firmware Intern *Palo Alto, CA*

- Enhanced future firmware reliability by implementing tooling PC-Lint for static code analysis on Renesas microcontroller
- Assessed non-volatile memory usage across all body control modules by introducing CAN logging for EEPROM usage
- Improved vehicle performance by fixing various bugs and implementing features related to sensor calibration and interfacing (mirror heaters, trailer brake sensors, overhead lights, door actuation)

Garmin **May 2023 - Aug 2023**
Flight Control Systems Firmware Intern *Olathe, KS*

- Accelerated aircraft control module testing by writing a Python SIL simulation for vertical navigation and altitude tracking
- Optimized flight control module performance by developing HIL unit tests in C within an RTOS environment

University of Texas at Austin **Aug 2024 - Present**
Operating Systems TA *Austin, TX*

- Enhance understanding of OS concepts for 70+ students through effective instruction
- Deliver 1-hour special topic lectures on bootloaders, microkernels, and memory allocators

ACTIVITIES AND PROJECTS

Longhorn Racing Solar Car **Sept 2021 - Present**
Platform Team Lead

- Improved robustness of on-vehicle firmware updates by developing a UART/CAN bootloader (custom linker script and flash memory driver)

Controls Team Lead

- Programmed in an RTOS context for our platform of STM32 microcontrollers (display, motor controller)
- Managed a team of 12 through a traditional software workflow with Git/GitHub

UT Embedded Systems Lab Design Competition - 3rd Place **Apr 2024**
Embedded Software Engineer

- Collaborated with a team of 4 to develop a GameBoy emulator on a TM4C microcontroller
- Designed a circuit board in KiCAD and worked with protocols such as SPI/QSPI and 16-bit parallel

UT Real-Time Operating Systems Graduate Project **Apr 2025**
Embedded Software Engineer

- Interfaced with a Wifi-Ethernet bridge chip and wrote MAC, IP, UDP, ICMP, ARP, and DHCP drivers from scratch on a TI microcontroller
- Demonstrated functional ping and netcat implementations on an RTOS

Skills: C/C++, KiCAD, Kernel Development, Python, Verilog, Vitis HLS, Application Development for RTOS, Linux, Git, Docker, AWS, Java
Interests: Table Tennis, Music Production, Piano